TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

GRADE 10

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · BAHÇEŞEHİR · İSTANBUL · İSTANBUL GALATA UNIVERSİTESİ

1. The point E is on the AB side of the square ABCD.
If [BD]$\cap$[CE]=F and |AE|=|EF| then what is the value of $\frac{|DF|}{|EF|}$?

A) $2\sqrt{3}$ B) $\sqrt{3}-1$ C) $\frac{3\sqrt{3}-2}{2}$ D) $\frac{3\sqrt{2}}{2}$ E) $\frac{3\sqrt{3}}{2}$

2. Given $f: N \to N$

$$f(n) = \begin{cases} n+3 & if\ n\ is\ odd \\ n+5 & if\ n\ is\ even \end{cases}$$

According to this what is the sum of the digits of the value of $\underbrace{(fofofo\ldots of)}_{2024\ times}(1)$?

A) 23 B) 24 C) 25 D) 26 E) 27

3. Given $x^2+2x-1=0$ find the value of $x^4+\frac{6}{2}$?

A) 34 B) 35 C) 36 D) 37 E) 38

4. $A \subseteq B$

Given the set A is a subset of the set B.
If the sum of the number of the subsets of the sets A and B is 68, how many subsets of the set B do not contain the set A?

A) 16 B) 24 C) 32 D) 36 E) 48

5. Given the inequality $|2a-24|+|b-a| \leq 2$
How many different ordered pairs (a,b) are there in integers?

A) 1 B) 3 C) 5 D) 7 E) 8

6. The parallelogram ABCD is divided into 4 parallelograms by drawing parallel lines to the sides AB and BC.
If the area of three of these parallelograms is 2, 3 and 4, what is the sum of the values that the area of the parallelogram ABCD can take?

A) $\frac{117}{2}$ B) $\frac{175}{3}$ C) $\frac{223}{6}$ D) $\frac{237}{8}$ E) $\frac{419}{6}$

7. Veysel and Şengül is making a study programme together. Each day, Veysel is solving $\frac{3}{2}$ of the number of questions he solved in the previous day and Şengül is solving $\frac{1}{4}$ of the number of questions she solved respectively in the previous day.Since they solved an equal number of questions on the 6th day of the programme, on the first day, how many times the number of questions solved by Şengül is the number of questions solved by Veysel?

A) 3^5 B) $2 \cdot 3^5$ C) $4 \cdot 3^5$ D) 6^5 E) 12^5

8. If then what is the $x^6 - x^5 + x^4 - x^3 + x^2 - x + 1 = 0$ value of $x^{50} - 3x^{35} + x$?

A) 1 B) 7 C) 17 D) 27 E) 3

9. Given the polinomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ if $P(1) = 10$, $P(2) = 20$, $P(3) = 30$ then find the value of $\frac{P(6) + P(-2)}{10} = ?$

A) 18 B) 52 C) 43 D) 53 E) 13

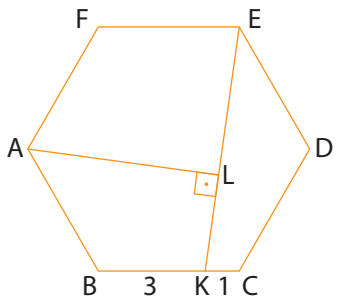
10.

				M				
M	A	T	E	M	A	T	İ	K
	A	T	E	M	A	T	İ	
M	A	T	E	M	A	T	İ	K
				M				

In the table above, the word MATEMATİK will be obtained by moving from one square to the other square from a common corner or a common edge.
In how many ways can this be done?

A) 1120 B) 1144 C) 1168 D) 1240 E) 1252

11.



In the regular hexagon ABCDEF given above

$[EK] \perp [AL]$, $|KC| = 1$ and $|BK| = 3$ if $|AL| = \frac{a\sqrt{b}}{c}$

where a, b and c are relatively prime natural numbers.

What is the value of the sum a+b+c?

A) 26 B) 28 C) 29 D) 32 E) 34

12. What is the remainder when 2023^9+2023^3 is divided by 2023^2+2024 ?

A) 1 B) 2 C) 3 D) 4 E) 5

13. a, b and c are relatively prime positive integers and $x = \frac{a}{b}$

If $x + \sqrt{x \cdot (x+3)} + \sqrt{x \cdot (9x+1)} + \sqrt{(x+3)(9x+1)} = 3$

then what is the value of a+b?

A) 17 B) 18 C) 19 D) 20 E) 21

14. Let A be the set of all functions which satisfy the followings

$$f: R \to R, \ f(f(x)) = x^2 \ and \ f(0) = 0$$

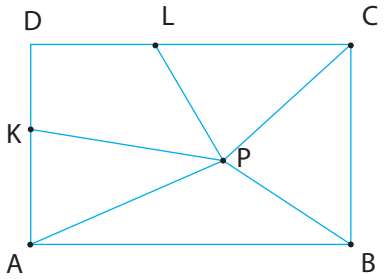
For the function in the set A how many different value can f(1) take?

A) 17 B) 18 C) 19 D) 20 E) 21

15. For a polinomial P(x) whose coefficients are all integers satisfying P(17)=10 and P(24)=17, the equation P(n)=n+3 satisfies for only two different integers n_1 and n_2.
What is the value of the product $n_1.n_2$?

A) 340 B) 348 C) 386 D) 400 E) 418

16.



In the figure above the points K and L are given on the sides AD and DC of the rectangle ABCD respectively and the point P is given inside of the rectangle. By connecting the point P with the points A, B, C, L, D and K through straight lines, the rectangle is divided into 5 regions whose areas are equal to each other.
According to this what is the value of the sum

$$\frac{|AK|}{|KD|} + \frac{|DL|}{|LC|} ?$$

A) $\frac{5}{2}$ B) $\frac{10}{3}$ C) $\frac{11}{2}$ D) $\frac{12}{5}$ E) $\frac{17}{3}$

17. Given the reel numbers x and y if $\left(2x+\sqrt{1+4x^2}\right)\left(3y+\sqrt{1+9y^2}\right)=1$
then find the value of $(2x+3y)^2$.

A) 0 B) 1 C) 2 D) 3 E) 4

18. Given pozitive reel numbers x, y and z if

$$x+y+z=20$$
$$\sqrt{256-x^2}+\sqrt{81-y^2}+\sqrt{16-z^2}=21$$

Then find the value of $\frac{x+y}{z}$?

A) 6,25 B) 6,5 C) 6,75 D) 7 E) 7,5

19. For all reel numbers $a_1, a_2, a_3, \dots$, an and $b_1, b_2, b_3, \dots$, bn the inequality

$(a_1 \cdot b_1 + a_2 \cdot b_2 + \dots + a_n \cdot b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2)$ holds.

According to this what is the greatest value of the reel number $\sqrt{x-1} + \sqrt{2x-5} + \sqrt{15-3x}$

A) 5 B) $\sqrt{26}$ C) $3\sqrt{3}$ D) $2\sqrt{7}$ E) $\sqrt{29}$

20. In how many different ways can 16 identical balls be distributed in 6 different boxes, provided that each box contains an odd number of balls?

A) 252 B) 244 C) 240 D) 210 E) 180

21. Given $f(x) = f(x-1) + 3x(x+1)$ and $f(10) = 0$ find the value of $f(1)$.

A) −1315 B) −1314 C) −1316 D) −1318 E) −1312

22. Given $x = 1 + \sqrt[4]{3} + \sqrt[4]{9} + \sqrt[4]{27}$ what is he value of $\left(\frac{2}{x} + 1\right)^{16}$?

A) 81 B) 108 C) 128 D) 243 E) 256

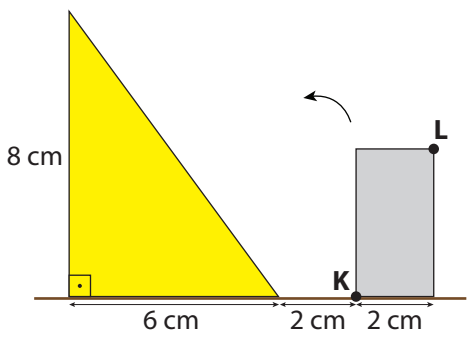
23. $\lfloor a \rfloor$ is defined as the greatest integer which is not greater than a.
Howmany different reel value of a are there which satisfies the equation $\lfloor a \rfloor \cdot (a - \lfloor a \rfloor) = 2024 \cdot a$?

A) 1 B) 2 C) 3 D) 4 E) In infinite multiplicity

24. How many different reel solution of the equati $\sqrt{x+20} + \sqrt[3]{13-x} = 7$ are there ?

A) 0 B) 1 C) 2 D) 3 E) 4

25.



The right side lengths of the yellow right triangle in the figure are 6 cm and 8 cm. The short side of the grey rectangle drawn on the same ground as the triangle is 2 cm. When the rectangle is rotated 90° around point K in the direction of the arrow, point L of the rectangle lies on the hypotenuse of the triangle.
Since the distance between the triangle and the rectangle is 2 cm, what is the area of the region where the triangle and the rectangle intersect after the rectangle is rotated in cm^2?

A) $\frac{3}{2}$ B) $\frac{3}{4}$ C) $\frac{5}{2}$ D) 2 E) $\frac{7}{4}$